

Woosung Song

CONTACT INFORMATION

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EDUCATION

University of California, Davis, CA

Ph.D. Candidate in Statistics, 2021 - Present (Expected graduation: June 2026)

Advisor: [Professor Hans-Georg Müller](#)

- Recipient of the Julius Blum Award (highest Ph.D. academic award in the department)
- Recipient of the Peter G. Hall Award (highest Ph.D. research award in the department)

Seoul National University, Seoul, South Korea

B.S. in Statistics and Mathematical Sciences (Double Major), 2021

Advisor: [Professor Hee-Seok Oh](#) & [Professor Taesung Park](#)

- Grade: Summa cum laude

RESEARCH INTEREST

Metric geometry and statistics; Manifold learning; Latent space representation; Optimal transport; Functional data analysis; Nonparametric statistics; Causal inference; Transfer learning; Brain networks

PUBLICATIONS & PREPRINTS

1. Choi, C.[†], **Song, W.**[†], Müller, H.-G., & Park, B. U. (2026). Additive Fréchet regression of random objects, *Under Major Revision, The Annals of Statistics*. [†]Equal Contribution
2. **Song, W.**[†], Zhou, H.[†], Zhou, Y.[†] & Müller, H.-G. (2026). Non-Euclidean data analysis with metric statistics, *Harvard Data Science Review, Accepted*. [[paper](#)] [[code](#)]
3. **Song, W.** & Müller, H.-G. (2026). ADOPT: Additive optimal transport regression, *AISTATS, Accepted*. [[paper](#)][[code](#)]
4. **Song, W.** & Müller, H.-G. (2026). Inference for dispersion and curvature of random objects, *Journal of the American Statistical Association*, 121(553), 729-740. [[paper](#)][[code](#)]
 - AISTATS 2026 Journal-to-Conference Track
 - Student Paper Award Finalist by ASA Nonparametric Statistics Section, JSM 2024
5. **Song, W.**, Oh, H.-S., Cheung, K. & Lim, Y. (2024). Multi-feature clustering of step data using multivariate functional principal component analysis, *Statistical Papers*, 65(4), 2109-2134. [[paper](#)][[code](#)]
6. Kim, H.[†], **Song, W.**[†], Choo, W., Lee, S., ..., Park, T., & Jang, J.-Y. (2023). Development, validation, and comparison of a nomogram based on radiologic findings for predicting malignancy in intraductal papillary mucinous neoplasms of the pancreas: An international multicenter study, *Journal of Hepato-Biliary-Pancreatic Sciences*, 30(1), 133-143. [[paper](#)]

7. Kang, J., Lee, C., **Song, W.**, Choo, W., ..., Park, T., & Jang, J.-Y. (2020)
Risk prediction for malignant intraductal papillary mucinous neoplasm of the pancreas:
logistic regression versus machine learning,
Scientific Reports, 10, 20140. [[paper](#)]

TALKS

ADOPT: Additive optimal transport regression

- The 29th International Conference on Artificial Intelligence and Statistics (AISTATS), Tangier, Morocco. (May 2026)

Inference for dispersion and curvature of random objects

- AISTATS, Journal-to-Conference Track, Tangier, Morocco. (May 2026)
- Joint Statistical Meetings (JSM), Topic-contributed Paper Session, Portland, OR. (Aug 2024)
- Statistics in the Age of AI, George Washington University, DC. (May 2024)
- Princeton Machine Learning Theory Summer School, Princeton University, NJ. (Jun 2023)

Multi-feature clustering of step data using multivariate functional principal component analysis

- Fall Korean Statistical Society Conference, University of Seoul, South Korea. (Nov 2019)

HONORS & AWARDS

IMS Hannan Graduate Student Travel Award, <i>Institute of Mathematical Statistics</i>	2026
Peter G. Hall Award, <i>Department of Statistics, UC Davis</i>	2025
Best Presentation Award, <i>ASA Nonparametric Statistics Section</i>	2024
Student Paper Award Finalist, <i>ASA Nonparametric Statistics Section</i>	2024
Travel Award, <i>Princeton Machine Learning Theory Summer School</i>	2023
Julius Blum Award, <i>Department of Statistics, UC Davis</i>	2022
Excellent Tutoring Award, <i>Seoul National University</i>	2020
Dean's List, <i>College of Natural Science, Seoul National University</i>	2020
3 rd Prize, <i>Poster presentation, Fall Korean Statistical Society Conference</i>	2019

TEACHING EXPERIENCE

University of California, Davis Associate Instructor

- STA103: Applied Statistics for Business and Economics (Summer 2025)
(Overall Rating: 4.75/5.0)

Teaching Assistant

- STA010: Statistical Thinking (Fall 2021, Spring 2022)
- STA142A: Statistical Learning I (Winter 2022)
- STA141B: Data & Web Technologies for Data Analysis (Fall 2022)
- STA206: Statistical Methods for Research I (Fall 2022)
- STA100: Applied Statistics for Biological Science (Winter 2023)
- STA013: Elementary Statistics (Spring 2023)
- STA103: Applied Statistics for Business and Economics (Spring 2025)
- STA104: Nonparametric Statistics (Fall 2025)
- STA138: Categorical Data Analysis (Winter 2026)

MENTORING

Muqing Cui (now Ph.D. student at UC Davis) Project: Fréchet variance process	2023 - Present
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SERVICE	Referee Journal of the American Statistical Association, Biometrika, Harvard Data Science Review	
	Conference Organization Chair, “Functional Data Analysis: Regression, Latent Structure, and Inference” session, Joint Statistical Meetings, Boston, MA. (Aug 2026)	
PROFESSIONAL EXPERIENCE	Capital One, Plano, TX Data Science Ph.D. Intern	Jun 2024 - Aug 2024
	Developed explainable AI (XAI) models to estimate risk of account-level auto loans, ensuring transparency and optimized performance.	
	Republic of Korea Army, South Korea Sergeant	Feb 2016 - Nov 2017
SOFTWARE	frechet Statistical Analysis for Random Objects and Non-Euclidean Data (R package on GitHub).	
REFERENCES	Dr. Hans-Georg Müller Distinguished Professor, Department of Statistics, University of California, Davis	✉ hgmueLLer@ucdavis.edu
	Dr. Jane-Ling Wang Distinguished Research Professor, Department of Statistics, University of California, Davis	✉ janelwang@ucdavis.edu
	Dr. Alexander Aue Professor, Department of Statistics, University of California, Davis	✉ aaue@ucdavis.edu